Factors attributed to a Games Genre Defining Success

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1 Overview

This study investigates what makes certain video games "genre-defining" among the vast catalogue of titles on Steam. By applying Principal Component Analysis (PCA) to a comprehensive dataset of 51,701 games, I identified key success metrics that explain 79.63% of variance in game performance, with the first principal component capturing 61.37% of variance across review scores, player counts, and ownership metrics which I used as a "Success Metric". K-means clustering identified distinct game success patterns, revealing that market saturation has reduced the likelihood of games achieving genre-defining status. Analysis of top games by genre showed that landmark titles like PUBG, Garry's Mod and Rust established genre conventions early when competition was lower, maintaining influence despite their age. These findings demonstrate that genre-defining success is primarily determined by innovation, versatility, Audience Balance, and cultural impact rather than release year.

2 Introduction

Context and motivation This study falls within data analysis and dimensionality reduction, specifically applying Principal Component Analysis to investigate what makes certain video games genre-defining. Despite the explosion in game releases, only a select few achieve genre-defining status, making it crucial to identify what factors truly distinguish these standout titles from the vast majority that fade into obscurity. I examine the paradox of declining game success despite increased production and technological advancements, using Steam which provides large volumes of data to explore. By combining Cluster and temporal analysis I have extracted patterns such as the impact market saturation has on game success over time and the factors that contribute to a genre defining success.

Previous work Many works have dived into this important question such as Van Dreunen [4] proposing the "Play Pendulum" theory, which argues that the video game industry alternates between periods of content innovation and distribution. While Li and Guo [11] conducted regression analysis on video game datasets to investigate factors influencing game success, finding significant correlations between commercial performance and game genres, user ratings; thematic content. And Finally Apperley [1] critiques conventional video game genre classifications for organizing games by aesthetic and narrative similarities to older media rather than by patterns of player interaction. But my analysis comes to similar conclusions differently by applying dimensionality reduction to apply the convention of top game success metrics into a singular applicable dimension and use that to determine the key factors of a video games' genre defining success while applying k-means to examine the implications of market saturation.

Objectives

- Identify the key factors that contribute to a games' success on Steam using dimensionality reduction techniques.
- Investigate patterns among genre-defining games and what distinguishes them from other titles.
- Analyse how market saturation affects the potential for new games to achieve genre-defining status.

3 Data

Data provenance The datasets used in this analysis were created by Newbieindiegamedev, they compiled the steam data through web scraping. I acquired this data by cloning the repository from GitHub [9] "steam-insights" to process and analyse on my own machine through VS-Code.

Data description The steam-insights dataset consists of multiple csv files, games.csv, categories.csv, genres.csv, reviews.csv, steamspy_insights.csv and tags.csv. After preprocessing and merging, the final data frame contains 51,701 rows and 14 columns with the key dimensions, 'release_year', 'positive', 'review_score', 'negative', 'total', 'concurrent_users_yesterday', and 'owners_average', 'tag' and 'genre'.

Data processing Data processing involved several steps: loading CSV files with pandas using appropriate encoding to handle special characters [10]; converting columns to proper data types (string, int, float); transforming genre/tag columns into lists; extracting release_year from datetime-formatted release_date; removing rows with missing values; separating paid and free games while filtering for relevant genres; and finally merging these frames on app_id. This process left me with a clean dataset with complete information across all dimensions, providing both numeric metrics for quantitative analysis and categorical data for qualitative exploration of genre-defining characteristics.

4 Exploration and analysis

Principal Component Analysis (PCA) was applied to multiple success metrics for interpretable components, providing a more comprehensive understanding of the dataset rather than analysing metrics in isolation. Given the heavily skewed nature of the data in Table 1, represented by Positive, negative, total reviews and game owners in the thousands to millions and Review Scores ranging from 0 to 10, a log transformation was applied to normalise the metrics. This step ensures comparability across variables and improves the readability of the PCA Plot (Figure 2), as recommended in [12].

release_year	positive	review_score	negative	total	concurrent_users	owners_average
2000	235403.0	9	6207.0	241610.0	11457	15000000.0
1999	7315.0	8	1094.0	8409.0	52	7500000.0
2003	6249.0	8	672.0	6921.0	82	7500000.0
2001	2542.0	8	524.0	3066.0	6	7500000.0
1999	22263.0	9	1111.0	23374.0	99	3500000.0

Table 1: Dimensions before log transformation to show skewness of data

The Scree plot (Figure 1) demonstrates a clear elbow at component 2, justifying the use of two principal components that capture 79.63% of total variance. Figure 2 presents PCA results using bootstrap sampling for statistical robustness, with games as blue points and feature vectors as red arrows. The close alignment of positive reviews, total reviews, and owners_average vectors along PC1 reveals their strong correlation and collective contribution to game success, validating PC1 as a comprehensive performance indicator rather than relying on isolated metrics.

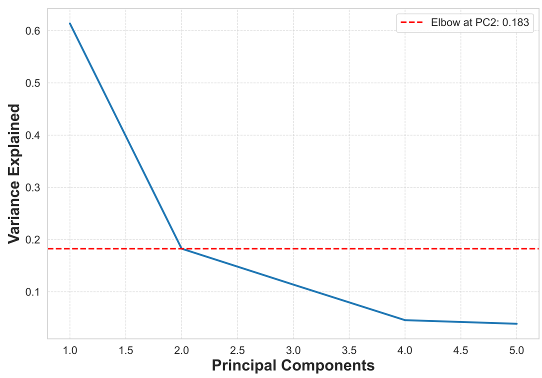

Figure 1: Scree plot to determine optimal number of principal components

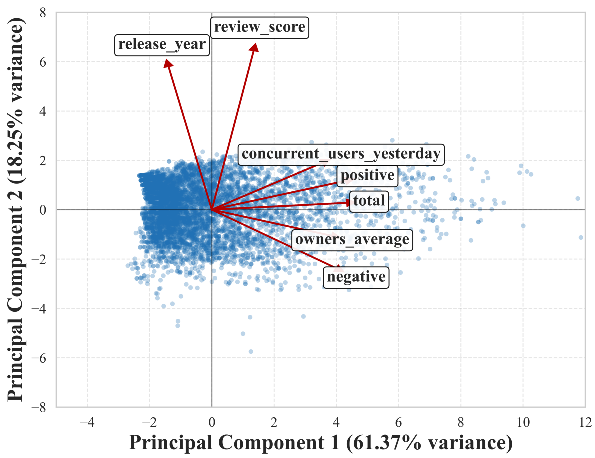

Figure 2: PCA Plot using three bootstrap samples, containing component feature vectors and their directions

Throughout my analysis I used PC1 as the primary "success metric" due to its explanatory power, capturing 61.37% of total variance. As Table 2 shows, PC1 features strong positive loadings for key performance indicators: total reviews (0.472), positive reviews (0.467), owners_average (0.421), and concurrent users (0.388). This creates a balanced metric integrating player engagement, commercial success, and critical reception into a single dimension. The negative loading for release_year (-0.149) suggests older games tend to perform better on this success metric, supporting the market saturation hypothesis examined later.

This dimensional reduction directly addresses my research questions by creating a unified success metric that enables fair comparison of games across genres and time periods; revealing patterns that would be obscured when examining metrics separately by visualising a preserved 79.63% of the data; identifying which aspects of game performance contribute most to genre-defining status through the loading values in Table 2. This data driven approach lays the foundation for identifying truly genre-defining games.

Feature	PC1	PC2
total	0.472	0.031
positive	0.467	0.127
negative	0.331	-0.257
owners_average	0.421	-0.116
concurrent_users_yesterday	0.388	0.208
release_year	-0.149	0.622
review_score	0.344	0.688

Table 2: PCA feature loadings across both principal components

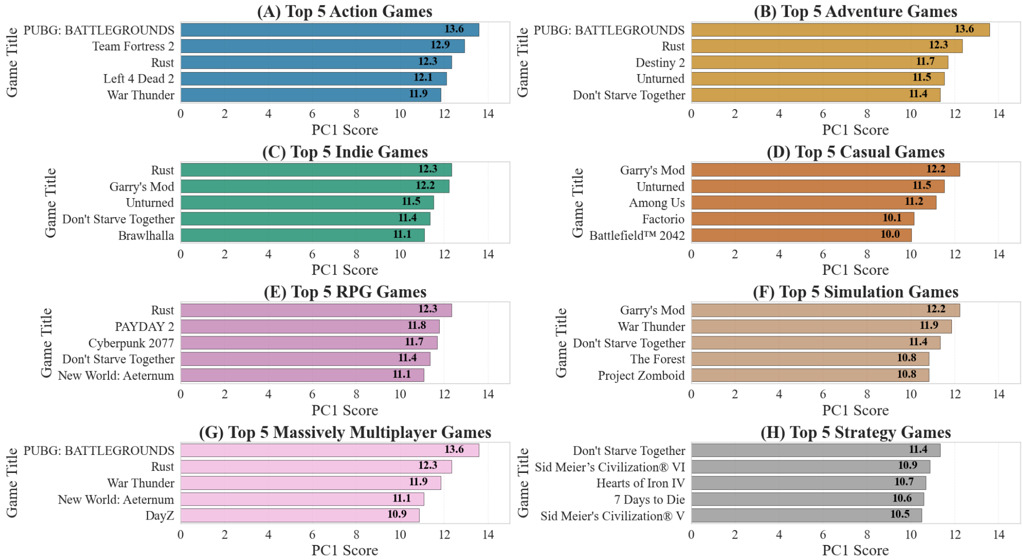

Figure 3: PCA Analysis of Top 5 games per Genre

The PCA analysis reveals consistent patterns among truly genre-defining games, with PUBG (13.6), Garry's Mod (12.2), and Rust (12.3) as seen in Figure 4(a) and Figure 3. First, innovation at the right time. PUBG "introduced the battle royale genre" [15] when the concept was in its infancy. As "The Game That Changed the Battle Royale Genre Forever" [13], PUBG didn't just build from prior concepts but revolutionised them surpassing predecessors such as DayZ [6]. Similarly, Garry's Mods' innovation of making "a game made from other games" [8], employing "innovative technology" with "passionate communities driving its growth forward" [16]. Meanwhile, Rust's focus on PvP reimaged survival gameplay [7].

PUBG's jump to mobile as showcased by Figure 4(B) with categories like "Remote Play on Phone" (8.2%) and "Remote Play on Tablet" (11.0%), "marked the beginning of a new era, where console-quality gaming experiences were no longer exclusive to dedicated gaming devices" [15], this showcased the dawn of a totally new mobile gaming experience not exclusive to Console or PC.

These games show significant interconnections, with Rust's creators acknowledging PUBG's influence on their approach to player conflict [3]. PUBG's PC1 score (13.6) outperforming DayZ (10.9) shown on Figure 3(G) demonstrates how it "changed the Battle Royale genre" [13], while Garry's Mod's incorporation of "Left 4 Dead" assets [14] benefited both titles as they made the top 10 caluclated by the Success Metric in Figure 4(A).

All three games achieve perfect audience balance. PUBG's tag distribution 4 showing transition from PC to Mobile.[15] (B). Garry's Mods' map system includes "simple maps to complex mini-games" [16]. Rust offering immediate objectives for new players while providing complex social systems for veterans [7]. These crucial features allows for broad appeal, creating entry points for casual players while satisfying competitive gamers.

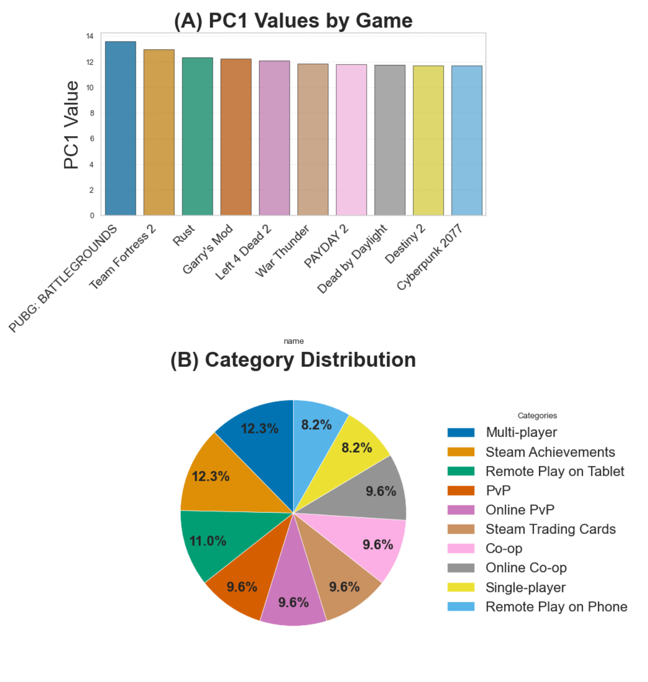

Figure 4: PCA Analysis of the top 10 games most common categories

The PCA data confirms these 4 factors - innovation, versatility, Audience Balance, and cultural impact – is what makes a genre defining game. But has market saturation affected the potential for new games to achieve genre-defining status, is it becoming less and less likely to achieve genre-defining success? And why is this?

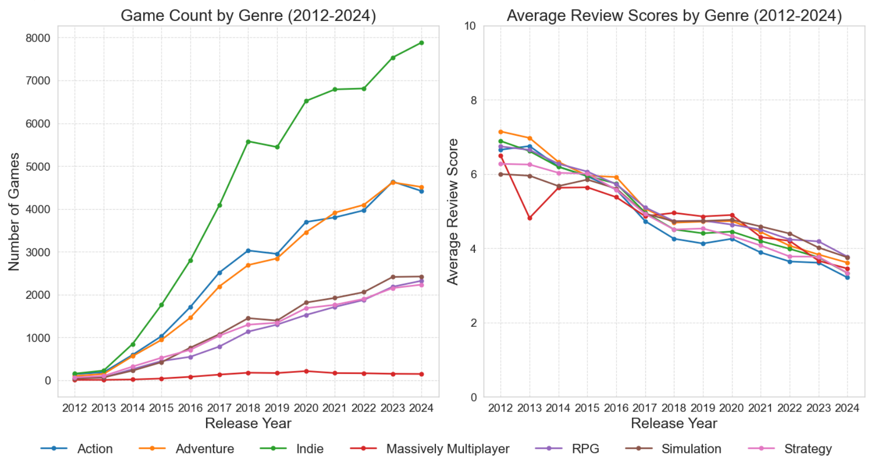

Figure 5: Temporal Analysis of game releases and average review scores per genre

The temporal analysis in Figure 5 reveals how genre-defining games relate to market saturation. Game releases across all genres have increased since 2013, with Indie titles showing growth exponentially, followed by Action and Adventure titles Figure 5(A). This creates a paradox where the more games that are getting released the less overall quality they become.

Figure 5 (B) confirms this statistically, showing declining average review scores across all genres since 2012, coinciding with the release surge and supporting the claim that there is a "decline in the overall quality of games in recent years" [2].

Clustering revealed patterns of market saturation, particularly in Indie Titles, linking well to the finding of Indie Dominance shown by Figure 5 (A). I extracted data from each cluster from Figure 6 where Cluster 3 has the lowest average PC1 score (-1.2) containing the most indie games (193), showing how over saturation reduces visibility and impact potential in other genres.

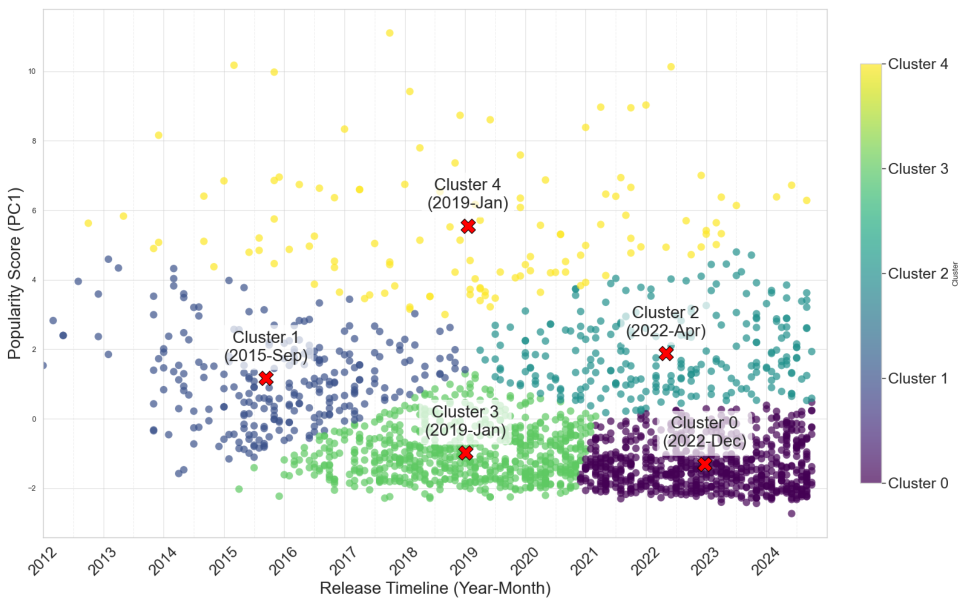

Figure 6: K-means clustering analysis of game releases by release timeline and popularity (PC1)

The high performing games are located in Cluster 4 with an average PC1 score of 5.55 with only 121 games present, showing the scarcity of impactful games opposed to Cluster 3 with 656 games. Cluster 4 having a median release date of April 2019 indicates diminishing opportunities for recent breakout success. Although genre-defining success remains possible, market saturation has significantly decreased the likelihood. Future revolutionary titles will require exceptional quality, innovation, strategic timing, and cross-platform versatility to overcome these barriers - qualities that PUBG, Garry's Mod, and Rust implemented, but in this bloated market the bar must be raised to achieve a similar impact to genre-defining games of the past.

5 Discussion and Conclusions

Summary of Findings PCA analysis revealed four key factors that represent genre-defining games: innovation at the right time, cross-platform versatility, audience balance, and cultural impact. PUBG, Garry's Mod, and Rust convey these principles, building upon existing concepts while introducing revolutionary elements that transformed their genres. Market saturation has significantly impacted the gaming landscape, with exponential growth in releases since 2013 coinciding with declining average review scores across all genres. K-means clustering confirmed this trend, showing high concentrations of poorly performing games (particularly in the indie category) in Clusters 0 and 3, while the high-performing Cluster 4 contains relatively few titles. The data strongly indicates that achieving genre-defining status has become increasingly difficult despite technological advancements, as the volume of releases increases.

Evaluation of own work: Strengths and Limitations The use of PCA and PC1 as a Success Metric effectively identified genre-defining games, with many articles consistently recognizing PUBG, Garry's Mod, and Rust as highly influential titles within their genres. However, while Principal Component 1 captured substantial variance in success metrics (61.37%), several critical data points were unavailable in the steamspy dataset. The absence of engagement metrics (playtime_average_forever, playtime_average_2weeks, playtime_median_forever, playtime_median_2weeks) and additional review indicators (metacritic_score, recommendations, steamspy_score_rank, steamspy_positive, steamspy_negative) limited the accuracy of my analysis. A more complete dataset would have enhanced the precision of my PCA. Additionally, being restricted to the Steam database prevented cross-platform comparison with influential titles like Minecraft and Fortnite [5], which would have provided valuable insights into genre-defining characteristics across different mediums of play.

Comparison with other related work My analysis showed many similarities with previous works. Apperleys [1] view on game genre classification is validated by my analysis, as my PCA (Figure 2) results are calculated by “player interactions” [1] such as reviews, owners and concurrent players (Table 2). Showing that genre-defining games like PUBG, Garry's Mod, and Rust (Figure 3) achieved success not purely through visuals. Van Dreunen's "Play Pendulum" theory [4] proposed that the video game industry alternates between innovation and distribution. My temporal analysis confirms this pattern, showing declining average review scores indicating declining innovation across genres since 2012 (Figure 5A) alongside the immense growth in releases (Figure 5B) indicating increased distribution. Unlike Li and Guo [11] findings of correlations between commercial performance and specific genres, my applied Principle Component Analysis (Figure 2) demonstrates that success is multidimensional - combining review quality and player engagement (Table 2). My research builds upon these previous works by employing dimensionality reduction techniques to create a success metric rather than relying on isolated performance indicators. This approach allowed me to identify key success factors while accounting for the correlations between player engagement and review quality, that the previous papers have examined separately rather than all dimensions as a whole..

Improvements and extensions If I had more time, I would have extended the database by web scraping relevant information for titles on other platforms to compare success patterns across all ways of playing. I would also fill in vital missing components from the datasets, particularly the playtime metrics and metacritic scores that were absent in the current data to enhance my PCA. Also wanted to label the top performing games on Figure 2 , this would have linked well with the further analysis of top games per genre in Figure 3. Would add in a table incorporating key statistics in each Cluser for Figure 6, this would improve readability. I also dived into the application of a regression model to predict genre-defining potential for new releases, but due to time constraints and space limitations wasn't able to incorporate it into this study.

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